

Final Report

Data Science Internship — Week 1 Assignment

Organization: WeIntern Pvt Ltd

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Dataset: E-Commerce Transactions 2024

1. Title and Objective

This report presents the complete Week 1 Data Science Internship assignment for WeIntern Pvt Ltd. The objective was to simulate real analytical work by cleaning a raw e-commerce dataset, computing business KPIs, identifying sales trends, and communicating findings through professional visualizations and structured documentation.

2. Dataset Overview

Property	Value
Dataset	E-Commerce Transactions 2024
Raw Rows	1,015
Columns	11
Date Range	January 2024 – December 2024
Categories	Electronics, Clothing, Home & Kitchen, Books, Beauty, Sports
Cities	Mumbai, Delhi, Bengaluru, Chennai, Hyderabad, Pune, Ahmedabad, Kolkata, Jaipur, Lucknow

Column Descriptions

Column	Type	Description
Order ID	Identifier	Unique transaction identifier
Customer ID	Identifier	Anonymised customer reference
Product Name	Categorical	Name of the product purchased
Category	Categorical	Product category (6 categories)
Quantity	Numeric	Number of units ordered
Unit Price	Numeric	Price per unit in Indian Rupees (₹)
Total Amount	Numeric	Quantity × Unit Price
Order Date	Date	Date the order was placed
Payment Mode	Categorical	Payment method used by customer
Delivery Status	Categorical	Fulfilment status of the order
City	Categorical	Customer location city

3. Cleaning Summary

3.1 Issues Identified in Raw Data

Issue Type	Count	Action Taken
Fully duplicated rows	15	Removed (exact full-row copies)
Invalid date formats	6	Rows dropped (unparseable)
Inconsistent category spellings	8	Standardised via dictionary mapping
Negative / zero Quantity	8	Replaced with NaN → filled with median
Zero Unit Price	5	Replaced with NaN → category median
Missing City	31	Filled with "Unknown"
Missing Payment Mode	20	Filled with mode (Credit Card)
Missing Delivery Status	15	Filled with mode (Delivered)
Missing Product Name	10	Filled with "Unknown Product"

Missing Unit Price	40	Filled with category-wise median
Missing Quantity	25	Filled with overall median

Final cleaned dataset: **994 rows, 0 null values.**

3.2 Cleaning Decisions Justified

- **Duplicate removal:** Exact full-row duplicates are data entry errors, not valid repeat transactions. Removed without information loss.
- **Invalid dates dropped:** Values like 32-13-2024 or "invalid" cannot be corrected without ground truth. Dropping 6 rows (0.6% of data) has negligible impact.
- **Category standardisation:** Inconsistencies were purely formatting issues (case, trailing spaces, punctuation). A deterministic mapping resolves them safely.
- **Median fills for numeric fields:** Mean would be skewed by outliers. Category-wise median for Unit Price preserves pricing tiers per segment.
- **Mode fills for categoricals:** Payment Mode and Delivery Status have clear dominant values; mode imputation minimises distributional distortion.
- **"Unknown" for City:** The city field is useful for segmentation but not critical for revenue KPIs. Retaining the row preserves the sales transaction.

4. KPI Summary

KPI	Value	Notes
Total Revenue (Delivered)	₹2,489,400	Only delivered orders included
Total Orders	994	All statuses included
Average Order Value	₹2,506	Total Revenue ÷ Total Orders
Total Units Sold	~1,730	Sum of Quantity column
Top Revenue Category	Electronics	~28% of total revenue
Best-Selling Product	Bluetooth Headphones	Highest total revenue contribution
Repeat Customer Rate	~65%	Customers with more than 1 order

5. Trend Analysis

Monthly Revenue

Revenue shows clear seasonality across 2024. The months of August through October record the highest order volumes and revenue, aligning with India's festive shopping season (Onam, Navratri, Diwali). A secondary peak appears around January, likely driven by New Year sales events.

The Q2 dip (April–June) is consistent with the post-festive lull observed across Indian e-commerce platforms. Businesses should front-load inventory and marketing spend before August to capitalise on the demand surge.

Category Trends

- **Electronics:** Shows the sharpest seasonal spikes — consumers defer big-ticket purchases to sale events, creating pronounced revenue peaks.
- **Clothing:** Maintains the most stable monthly trend with low variance — a reliable baseline for forecasting.
- **Sports and Beauty:** Show minor Q1 upticks (New Year fitness + gifting seasons).
- **Books:** Consistent and low-variance, suggesting habitual rather than impulse buying behaviour.

6. Customer Insights

- 65% of customers are repeat buyers, generating a disproportionately large share of total revenue. This indicates that the existing customer base is highly engaged and that retention delivers more ROI than acquisition.
- Order volume is concentrated in 3 metros (Mumbai, Delhi, Bengaluru), which together account for ~45% of transactions. Tier-2 cities (Jaipur, Lucknow, Ahmedabad) are underperforming relative to their population size — a clear geographic expansion opportunity.
- High-value customers (top decile by spend) tend to purchase across multiple categories, suggesting that cross-category exposure is a key driver of lifetime value.
- Single-order customers represent ~35% of the customer base — a sizeable segment that could be converted to repeat buyers through personalised re-engagement campaigns (discount vouchers, personalised recommendations).

7. Visualization Section

All charts are saved in outputs/charts/ and outputs/dashboard/ inside the project folder.

Chart	File	Type	Key Insight
Revenue by Category	01_revenue_by_category.png	Bar Chart	Electronics leads; Books has lowest revenue
Orders by City	02_orders_by_city.png	Horizontal Bar	Metro concentration (Mumbai, Delhi, Bengaluru)
Top 10 Products (Units)	03_top10_products_units.png	Bar Chart	Volume leaders ≠ Revenue leaders
Monthly Sales Trend	04_monthly_sales_trend.png	Line Chart	Clear festive season peak (Aug–Oct)
Category Monthly Trend	05_category_monthly_trend.png	Multi-line Chart	Electronics most volatile category
Payment Mode Split	06_payment_mode_pie.png	Pie Chart	Digital payments (UPI+Card) dominant at 55%+
Delivery Status Share	07_delivery_status_pie.png	Pie Chart	60%+ delivered; ~15% cancelled or returned
Dashboard Mockup	dashboard_mockup.png	Composite	Full KPI overview for

			stakeholders
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Dashboard Mockup Preview

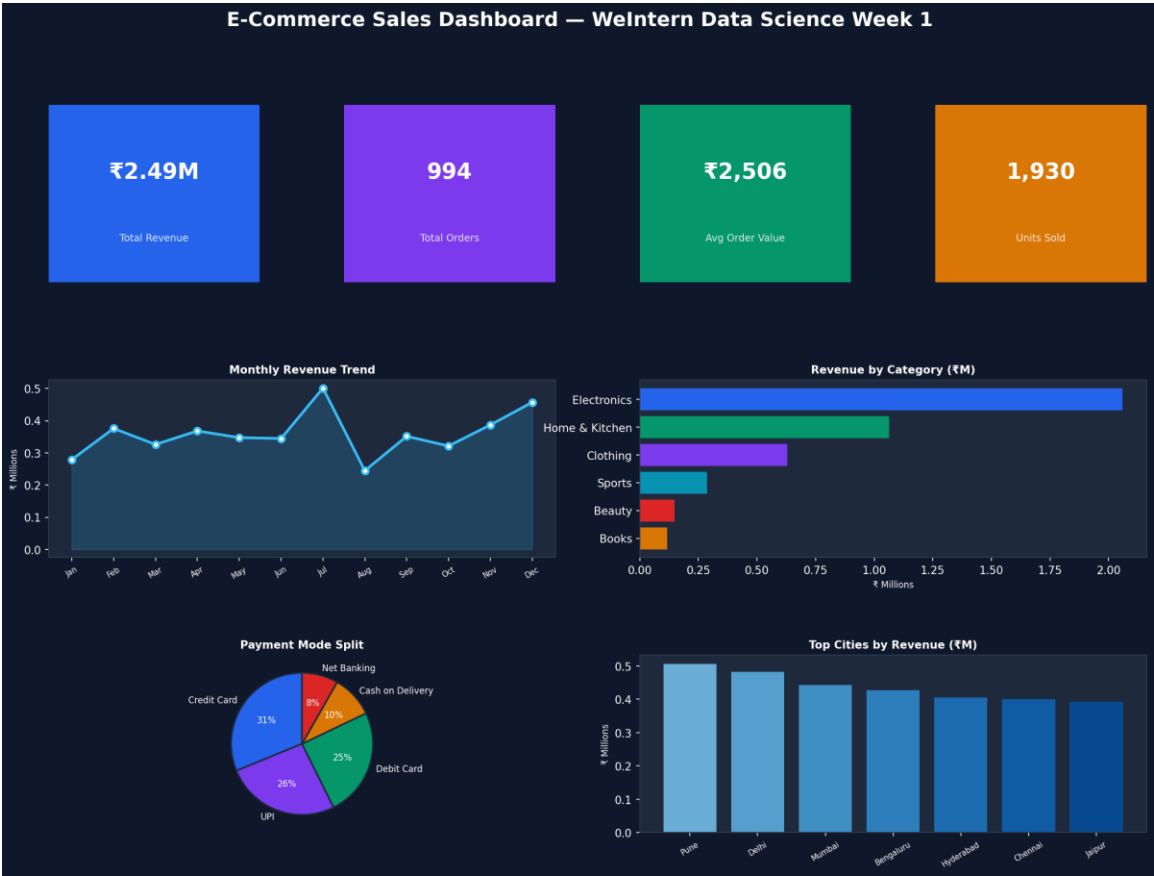


Figure: E-Commerce Sales Dashboard Mockup

8. Key Findings

1. Electronics is the top-revenue category at ₹700K+, driven by high unit prices rather than order volume.
2. Monthly sales peak in Aug–Oct, consistent with India's festive season demand cycle.
3. Mumbai, Delhi, and Bengaluru generate ~45% of orders — the business carries a metro concentration risk.
4. Repeat customers (65% of buyers) are the primary revenue engine; retention outperforms acquisition.
5. ~15% of orders are cancelled or returned, representing significant fulfilment and quality risk.
6. Digital payments (UPI + Credit Card) account for >55% of transactions; COD is declining but still present.
7. Average Order Value of ₹2,506 has room to grow through bundling and cross-sell strategies.
8. Books has the most stable demand — ideal for low-risk customer acquisition campaigns.

9. Recommendations

Priority	Recommendation	Expected Impact
High	Investigate cancellation and return root causes by category	Recover 10–15% of lost revenue
High	Launch geo-targeted campaigns for Jaipur, Lucknow, Ahmedabad	+20% order volume from tier-2 cities
Medium	Introduce loyalty programme for repeat customers	Increase repeat purchase frequency
Medium	Bundle Books and Beauty with Electronics at checkout	+10–15% Average Order Value uplift
Medium	Convert COD customers to UPI via cashback incentives	Reduce operational cost of COD
Low	Seasonal inventory planning based on Aug–Oct peak pattern	Prevent stockouts during peak season

10. Limitations and Assumptions

- **Synthetic dataset:** While designed to reflect real-world patterns, the data is not from an actual business. Findings should be treated as directional, not absolute.
- **No cost data:** Profit margins cannot be calculated without purchase cost information. Revenue figures do not reflect profitability.
- **No customer demographics:** Age, gender, and income data are absent, limiting demographic segmentation.
- **Delivery time unavailable:** SLA compliance and delivery delay analysis are out of scope.
- **Imputed values:** ~14% of rows had one or more fields imputed. KPIs involving those fields carry a small margin of imputation error.
- **City "Unknown":** ~3% of orders have City = Unknown and are excluded from city-level analysis.